

Installation and Programming of the DE-10 Agilex

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Installation

One time a get the license, the next step is install in the Farm (PC UBUNTU). First i download the instalator on Altera' page, We need to find the .run file, which is the Windows equivalent of the .exe file.

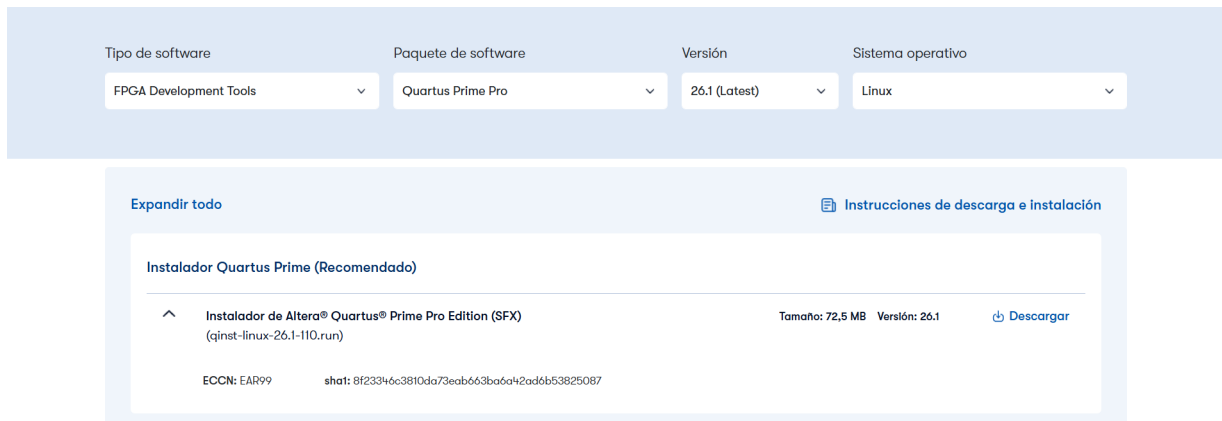


Figure 1. Altera download page.

Once the download is complete, we need to find the folder where our file is located.

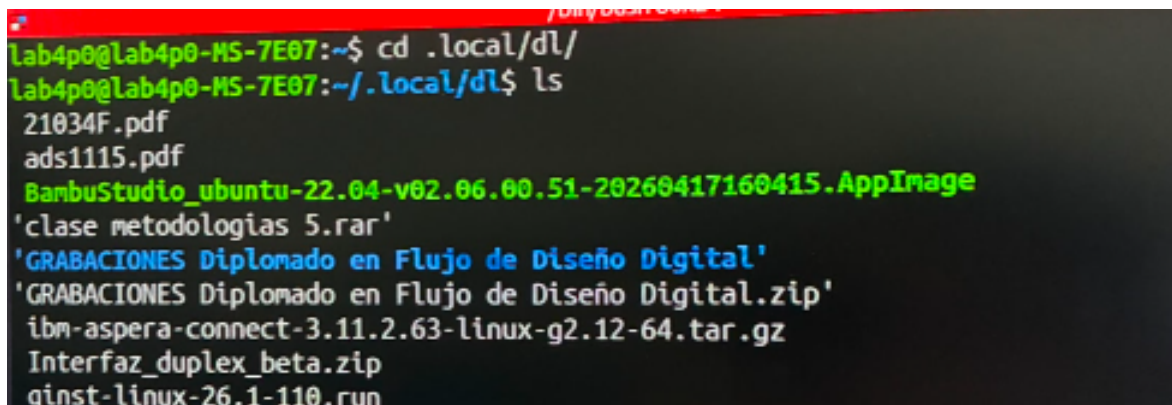


Figure 2. Linux terminal

For this, we use the 'cd' command to navigate to the file where we saved it, and in our case, the file was saved in the 'local/dl'. We also use the 'ls' command to verify that the file is indeed there. As we can see in Figure 2, the file 'qinst-linux-26.1-100.run' is indeed there, but the problem is that the file lacks executable permissions. (Executable files are highlighted in green).

```

Lab4p0@lab4p0-MS-7E07:~/local/dl$ chmod +x qinst-linux-26.1-110.run
Lab4p0@lab4p0-MS-7E07:~/local/dl$ ls
21034F.pdf
ads1115.pdf
BambuStudio_ubuntu-22.04-v02.06.00.51-20260417160415.AppImage
'clase metodologias 5.rar'
'GRABACIONES Diplomado en Flujo de Diseño Digital'
'GRABACIONES Diplomado en Flujo de Diseño Digital.zip'
ibm-aspera-connect-3.11.2.63-linux-g2.12-64.tar.gz
Interfaz_duplex_beta.zip
qinst-linux-26.1-110.run

```

Figure 3. Linux terminal part 2

For this we use the `chmod` command. This command is usually used to grant permissions; in this case, we are giving permission to make the file executable with the `"x"`. Then we use `ls`, and as we can see in the Figure 3, the file `qinst-linux-26.1-110.run` is now green, meaning it is executable.

```

Lab4p0@lab4p0-MS-7E07:~/local/dl$ bash qinst-linux-26.1-110.run
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing Intel Quartus Prime Pro Edition Installer 26.1 100%
Lab4p0@lab4p0-MS-7E07:~/local/dl$ ./qinst-linux-26.1-110.run
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing Intel Quartus Prime Pro Edition Installer 26.1 100%
Lab4p0@lab4p0-MS-7E07:~/local/dl$ sudo bash qinst-linux-26.1-110.run

```

Figure 4. Linux terminal part 3

Once the file is executable, we need to run it with the `bash` command. This executes a script file containing a list of commands. We can also use `./` to run the files. We can see the examples en the Figure 4.

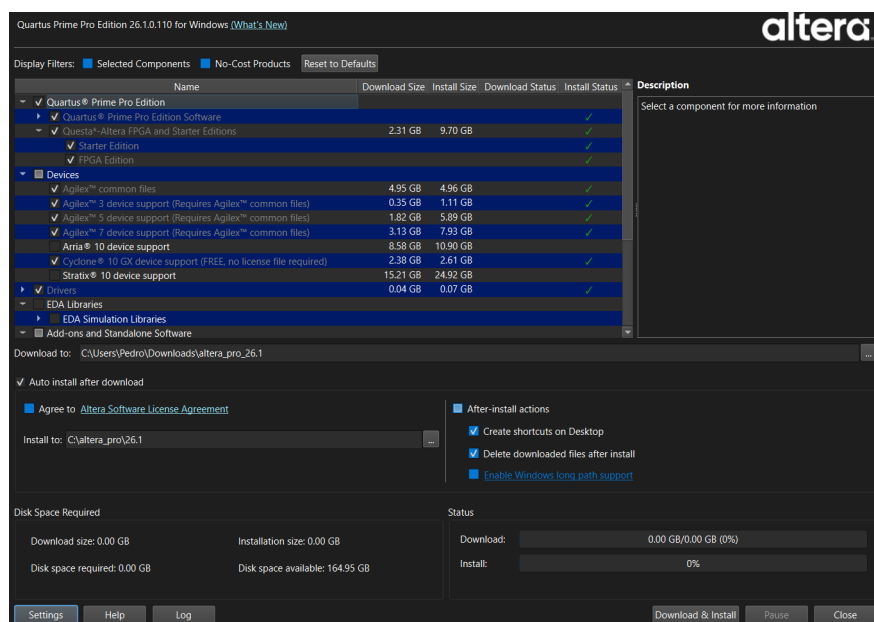


Figure 5. Quartus Prime Pro installer

Once you're in the Quartus installer, the most important thing is to select the Agilex 7 family, which is the chip in the Agilex DE-10. The other important thing is installing the "drivers" so that your device can be recognized by the FPGA.

We'll save it in the .config folder so the Quartus executable isn't so exposed to every user of the computer. Once the installation is complete, we'll locate the Quartus executable. We'll again use the "cd" and "ls" commands to find it and make sure it's in the directory where we saved it

```
lab4p0@lab4p0-MS-7E07:~$ cd .config/altera_pro/
lab4p0@lab4p0-MS-7E07:~/.config/altera_pro$ ls
26.1
lab4p0@lab4p0-MS-7E07:~/.config/altera_pro$ cd 26.1
lab4p0@lab4p0-MS-7E07:~/.config/altera_pro/26.1$ ls
devdata  licenses  pta      qemb      qsys      questa_fse  vds
devices  logs      qcore    qinit.sh  quartus    syscon
ip        niosv     qdrivers qprogrammer  questa_fe  uninstall
lab4p0@lab4p0-MS-7E07:~/.config/altera_pro/26.1$ bash qinit.sh
added /home/lab4p0/.config/altera_pro/26.1/quartus/bin to PATH environment
added /home/lab4p0/.config/altera_pro/26.1/qcore/bin to PATH environment
added /home/lab4p0/.config/altera_pro/26.1/questa_fse/bin to PATH environment
```

Figure 6. Linux terminal part 4

```
added /home/lab4p0/.config/altera_pro/26.1/quartus/bin to PATH environment
Updated PATH: /home/lab4p0/.config/altera_pro/26.1/quartus/bin:/home/lab4p0/.config/altera_pro/26.1/qcore/bin:/home/lab4p0/.config/altera_pro/26.1/questa_fse/bin:/home/lab4p0/.config/altera_pro/26.1/niosv/bin:/home/lab4p0/.config/altera_pro/26.1/vds/bin:/home/lab4p0/.config/altera_pro/26.1/syscon/bin:/home/lab4p0/.config/altera_pro/26.1/questa_fe/bin:/home/lab4p0/.config/altera_pro/26.1/questa_fse/bin:/home/lab4p0/bin:/home/lab4p0/bin:/home/lab4p0/bin:/home/lab4p0/.local/bin:/home/lab4p0/bin:/home/lab4p0/synopsys/02_tools/snps_container/2.1/bin:/home/lab4p0/synopsys/02_tools/fusto/bin:/home/lab4p0/synopsys/02_tools/vcs/X-2025.06-SP1-1/bin:/home/lab4p0/synopsys/02_tools/syn/X-2025.06-SP4/bin:/home/lab4p0/.local/bin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/home/lab4p0/altera_lite/25.1std/q
```

Figure 7. Linux terminal part 5

Before running the Quartus executable, we first run qinit.sh. This file updates the PATH environment variable, which appends the specific installation directories of Quartus Pro, Questa, and Synopsys tools to Linux's search list. This setup ensures that the system can instantly locate and run these EDA tools from any directory, while allowing the core software to seamlessly communicate with its internal synthesizers and compilers during the FPGA development workflow. We can see the results in the Figure 7

```
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite$ lf
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite$ cd quartus/bin/
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite/quartus/bin$ bash quartus
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite/quartus/bin$ bash quartus
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite/quartus/bin$ cd ..
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite$ cd ..
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite$ sudo mkdir licencias
[sudo] password for lab4p0:
lab4p0@lab4p0-MS-7E07:~/home/lab4p0/altera_lite/25.1309/altera_lite$ cp ~/.local/dl/LR-164130_licens
e.dat .
```

Figure 8. Linux terminal part 6

Because we couldn't find exactly where the Quartus executable was located, a new command, "lf", was used. The lf Command (short for List Files) launches a lightweight, terminal-based file manager that allows you to navigate directories, preview, and organize files through a text-based user interface. Heavily inspired by Vim, it utilizes a multi-column layout and keyboard shortcuts to provide a fast, efficient alternative to navigating the file system without needing a mouse or a heavy graphical user interface. (Sorry I dint' take a picture, later i will do a update).

Using this command, we can find the actual location of the file. It's located in the "quartus/bin" folder. The "quartus" executable, named quartus, is located there.

So we run it with bash and if it runs correctly we will see the interface in the Figure 9.

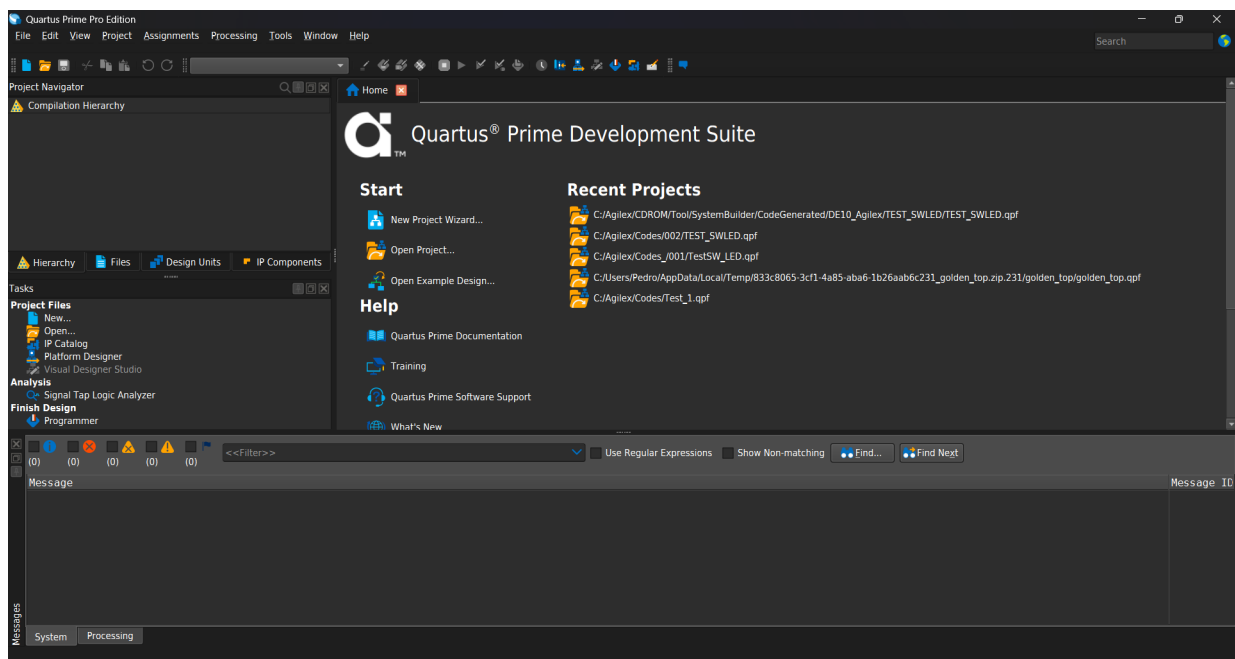
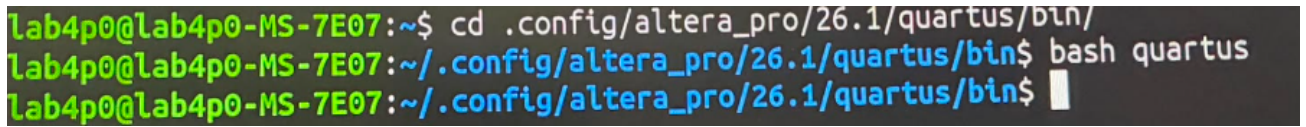


Figure 9. Quartus Prime Pro Software

WARNING: When you start Quartus Prime Pro, it will ask for a license, which is required to use the software. Therefore, it's recommended to create a folder, as shown in the Figure 8, using the "mkdir" command, where you are certain the license file is located.

Note that the "cp" command was used to copy the license file from "/local/dl" to "Licenses". This will allow us to move it to our trusted folder.

A screenshot of a Linux terminal window. The prompt is 'lab4p0@lab4p0-MS-7E07:~\$'. The first command entered is 'cd .config/altera_pro/26.1/quartus/bin/'. The second command is 'bash quartus'. The prompt changes to 'lab4p0@lab4p0-MS-7E07:~/ .config/altera_pro/26.1/quartus/bin\$' after the first command, and then to 'lab4p0@lab4p0-MS-7E07:~/ .config/altera_pro/26.1/quartus/bin\$' after the second command. A cursor is visible at the end of the second command line.

```
lab4p0@lab4p0-MS-7E07:~$ cd .config/altera_pro/26.1/quartus/bin/
lab4p0@lab4p0-MS-7E07:~/ .config/altera_pro/26.1/quartus/bin$ bash quartus
lab4p0@lab4p0-MS-7E07:~/ .config/altera_pro/26.1/quartus/bin$
```

Figure 10. Linux terminal part 7

Finally, in the Figure 10, we can see the two commands to run Quartus on the server PC.

- The first (cd) is to move to the folder with the executable.
- The second (bash) is to run it.

Programming

To begin, we must highlight that the programming for the DE10-Agilex and, for example, a DE1-SOC, is different. For instance, each requires different software: the DE1-SOC uses Quartus Prime Lite, and the DE10-Agilex uses Quartus Prime Pro. And visually the software programs are a little different, but overall they are very similar.

The first program we chose to program for the DE-10 Agilex was a program to turn on LEDs using switches. This way, we have a visual reference and can be sure that the board was programmed correctly.

In our initial attempts, we encountered some errors, for example, when assigning pins. Unlike the Quartus Prime Lite, we needed to use the correct I/O standard because the Agilex DE-10 is very strict regarding the voltage used. To ensure we identified the correct pins for the switches and LEDs, we used the Agilex DE-10 guide available on the Altera website.

Even so, we have a problem: even if we moved the pins and assigned them correctly, the Agilex still gave us a Power Management error. So we had to do a little research to find out what this error meant and how to fix it, since this error prevented the .SOF file from being generated.

Through our investigation, we discovered a specific functionality of this type of card from the Altera family; these cards have the following features: The explicit inclusion of VID (Voltage Identification) assignments and I/O constraints in the .qsf file is a mandatory requirement imposed by the Secure Device Manager (SDM) of the Intel Agilex 7 architecture. Unlike static-voltage academic platforms, Agilex employs the industrial PMBus protocol to dynamically govern the power loop via the LTC3888 controller. The omission of these geometric and electrical parameters is classified by Quartus Prime Pro as a critical design rule violation (DRC), disabling the generation of the executable bit-stream (.sof) to mitigate risks of thermal degradation or failures caused by floating pins in the silicon."

To solve this problem, further investigation was conducted. On the Altera website, there are examples of how to use the De-10Agilex in a resources section called CD-ROM. Here we can find a lot of information about the DE-10 Agilex; what interests us is a program called Systembuilder.

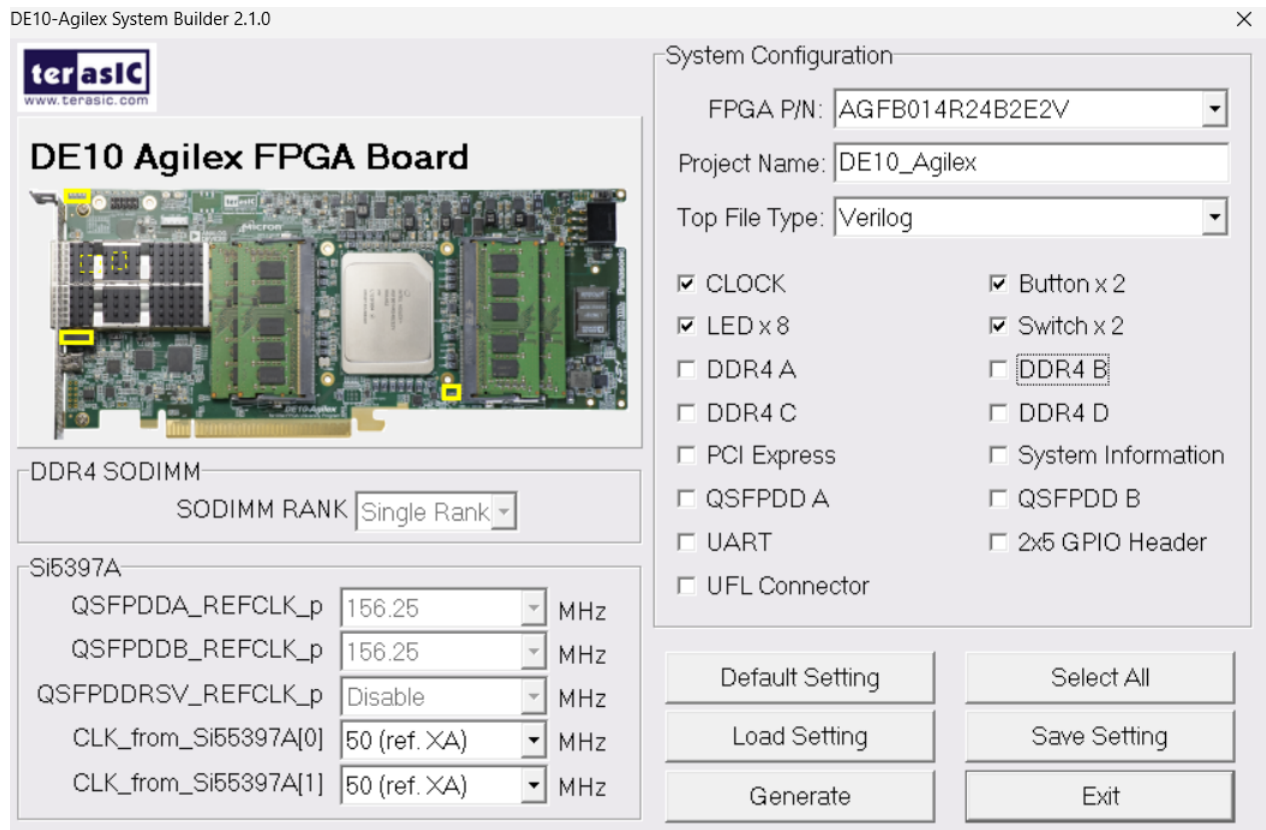


Figure 11. SystemBuilder user interface

With the systembuilder we can select the parts of the card we will use, and when we click generate it should give us a complete Verilog file with all the necessary files to run it in Quartus.

Even so, there was a problem, because what was done for an early version of Quartus had certain inconsistencies with the version of Quartus we were using (26.1). So we had to adapt it by changing the QSF file. The most significant changes were made to the electrical characteristics, due to the card's security system.

Once the changes are made, we'll have a template that will work for most cases, so all that's left is to program in the .V file generated by the systembuilder. There's a section in that file called "structural design" where we can write all the logic of our code, and then it should be able to load onto the card.